



# More Stellar Evolution



# Stellar Structure

- Stellar structure:
  - We know how to compute the structure of a star given its mass and composition
  - We also know how the composition changes due to nuclear reactions
  - Together this means we can compute the evolution of stars
    - Assuming HSE + thermal equilibrium maintained
  - We saw that polytropes can help us understand some types of stars

$$\frac{dP}{dM} = -\frac{GM}{4\pi r^4}$$

$$\frac{dr}{dM} = \frac{1}{4\pi r^2 \rho}$$

$$\frac{dL}{dM} = \epsilon$$

$$\nabla = \begin{cases} \nabla_{\text{rad}} & \text{if } \nabla_{\text{rad}} < \nabla_{\text{ad}} \\ \nabla_{\text{ad}} & \text{otherwise} \end{cases}$$

# Stellar Evolution

- Stellar evolution:
  - During some phases of evolution, we cannot neglect the time derivatives
    - Energy being released on very short timescales
    - Still assumes LTE
  - Need to solve the full set of PDEs
  - This is what MESA does

$$\ddot{r} = -\frac{GM}{r} - 4\pi r^2 \frac{\partial P}{\partial M}$$

$$\dot{e} + P \frac{\partial}{\partial t} \left( \frac{1}{\rho} \right) = \epsilon - \frac{\partial L}{\partial M}$$

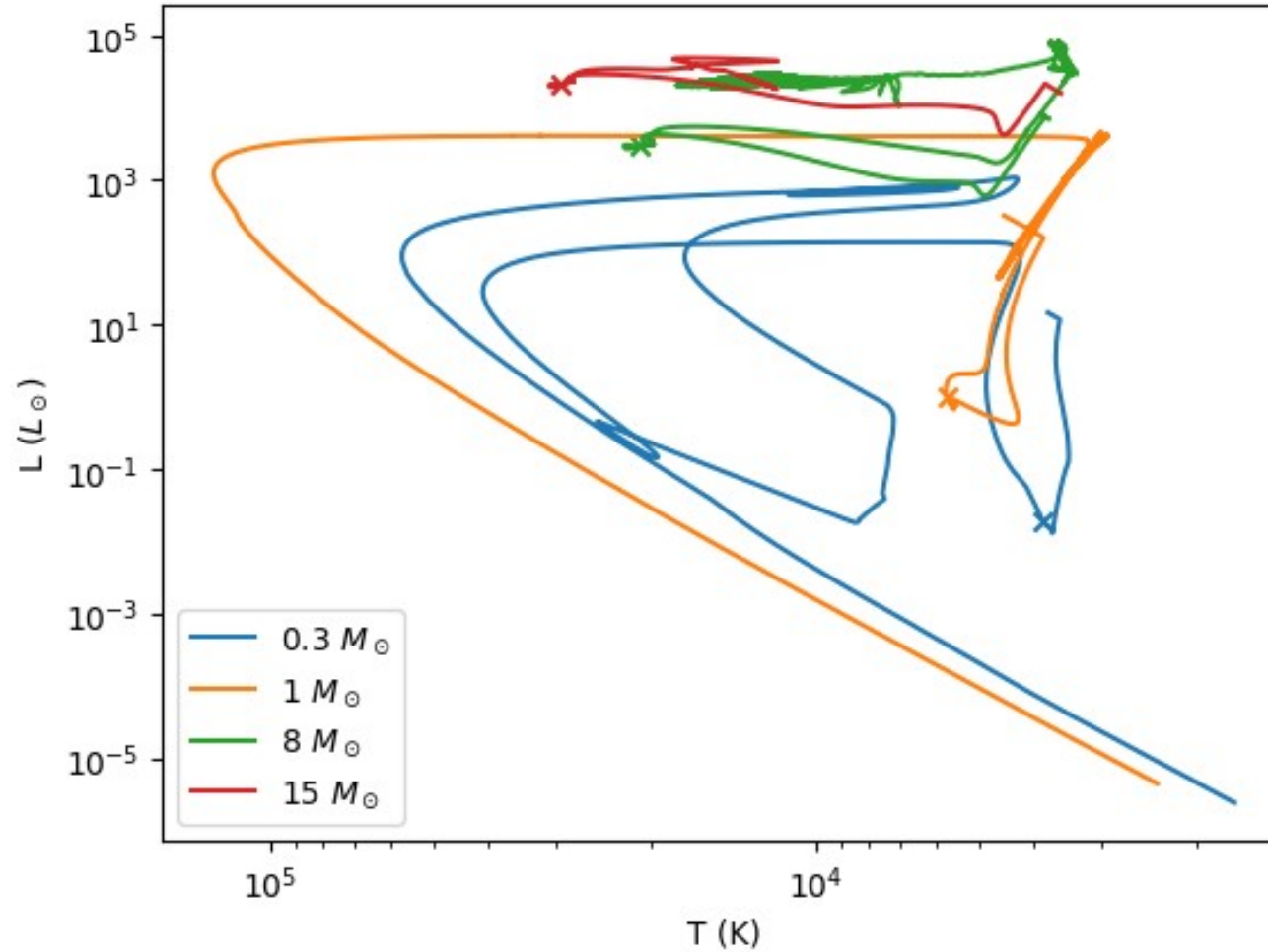
$$\dot{X}_k = f_k(\rho, T, X_j)$$

$$\frac{dr}{dM} = \frac{1}{4\pi r^2 \rho}$$

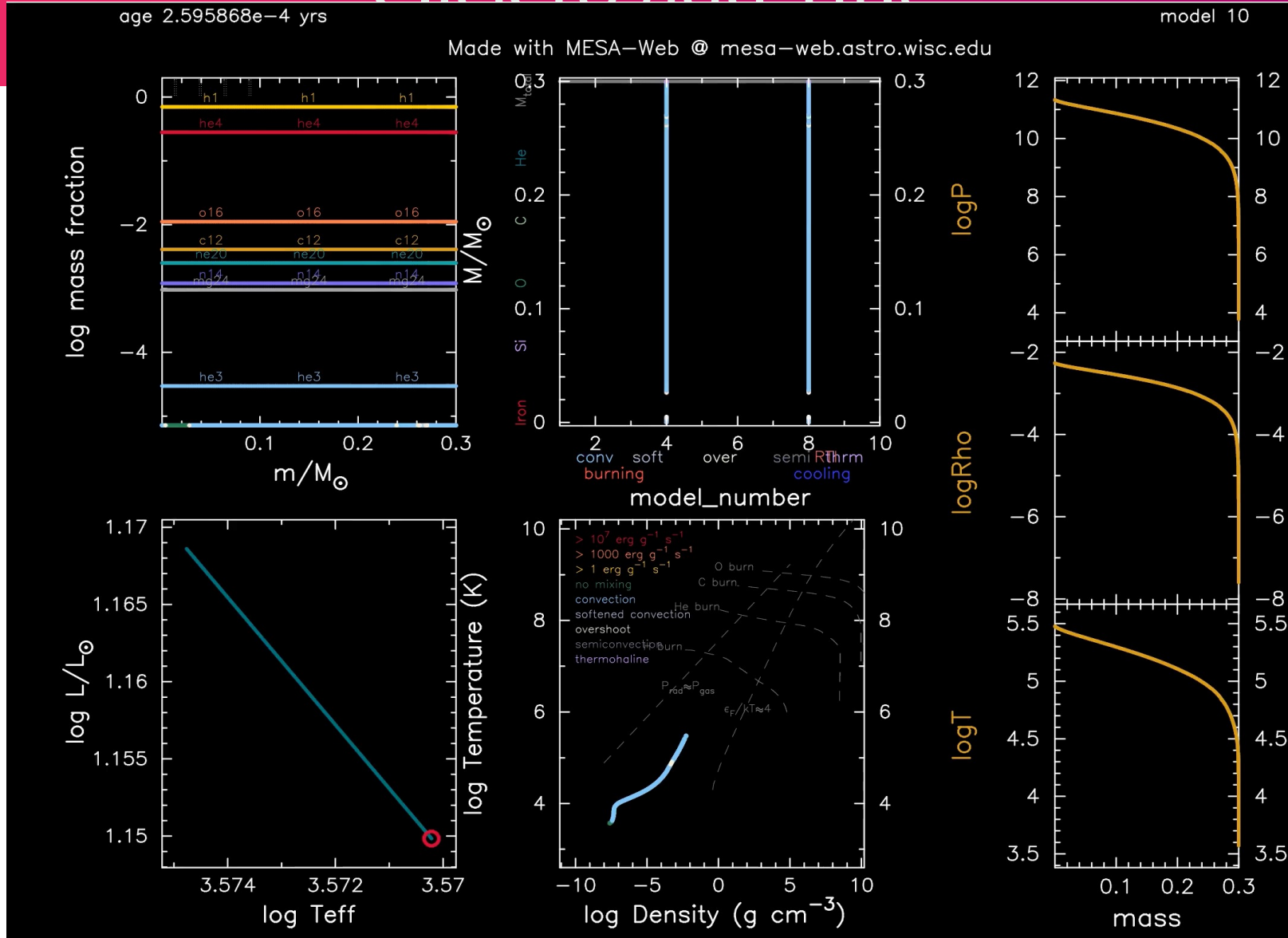
$$\nabla = \begin{cases} \nabla_{\text{rad}} & \text{if } \nabla_{\text{rad}} < \nabla_{\text{ad}} \\ \nabla_{\text{ad}} & \text{otherwise} \end{cases}$$

# Stellar Evolution

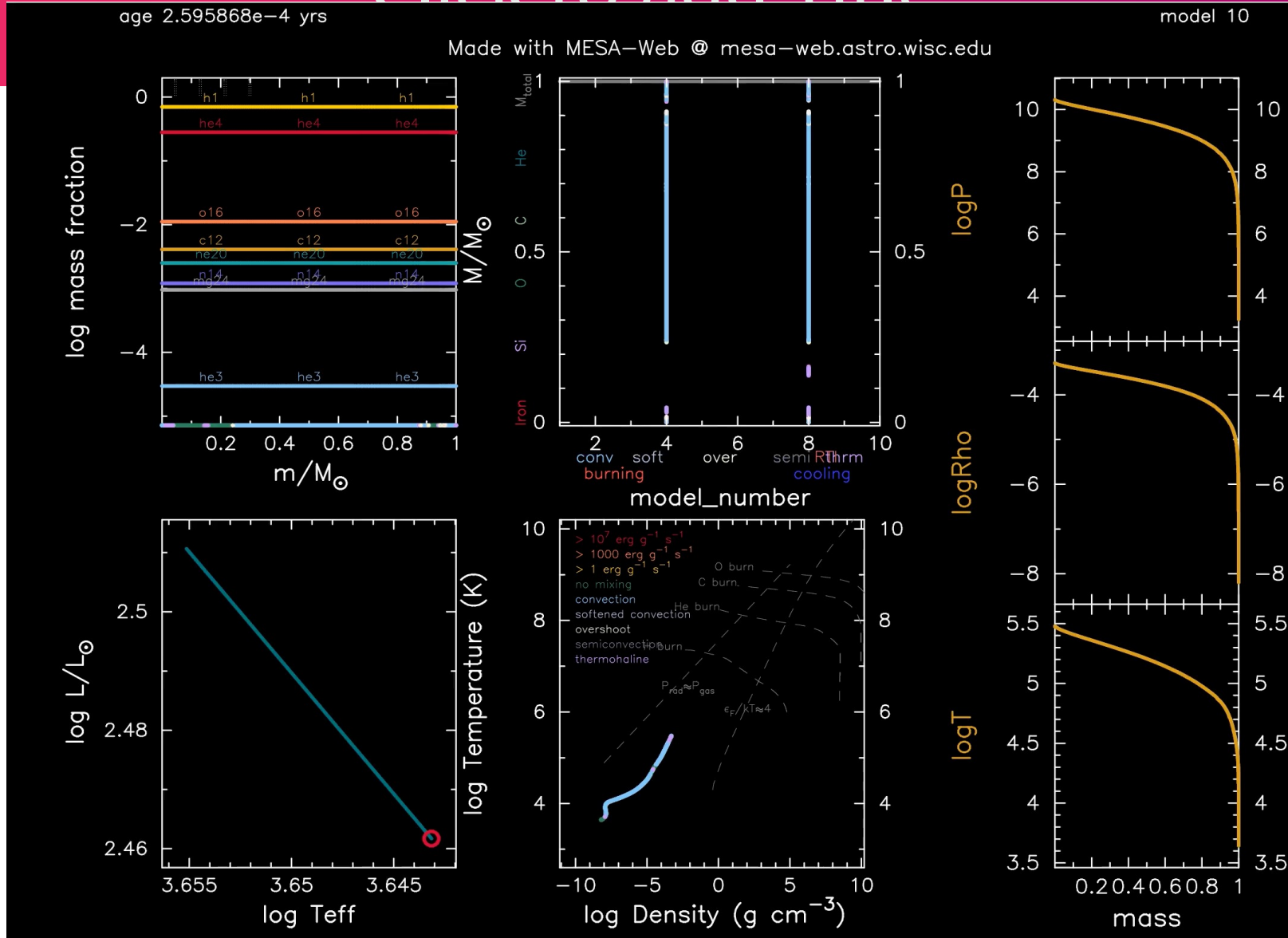
- Let's look at MESA output for 4 masses:  $0.3 M_{\odot}$ ,  $1 M_{\odot}$ ,  $8 M_{\odot}$ , and  $15 M_{\odot}$



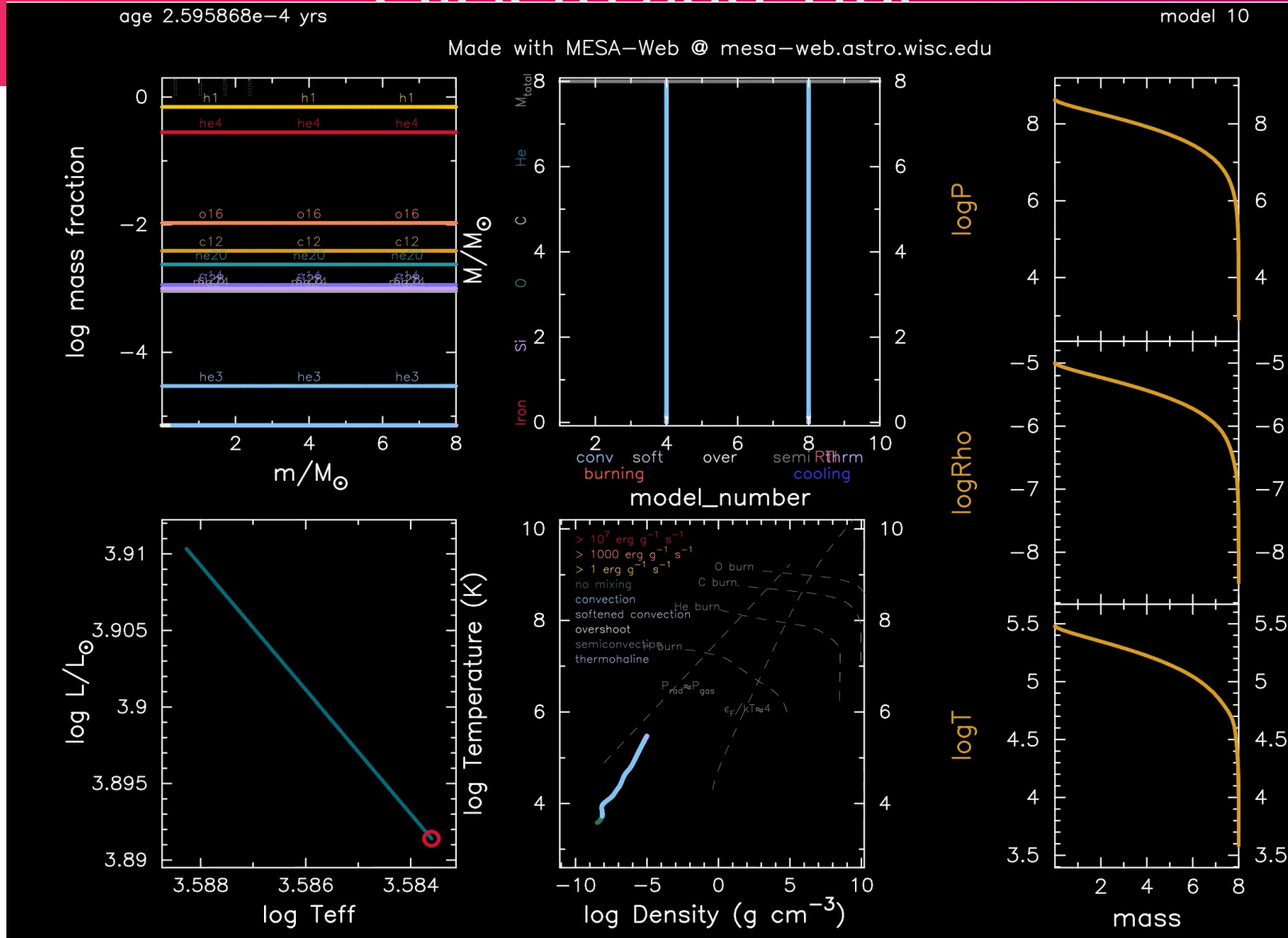
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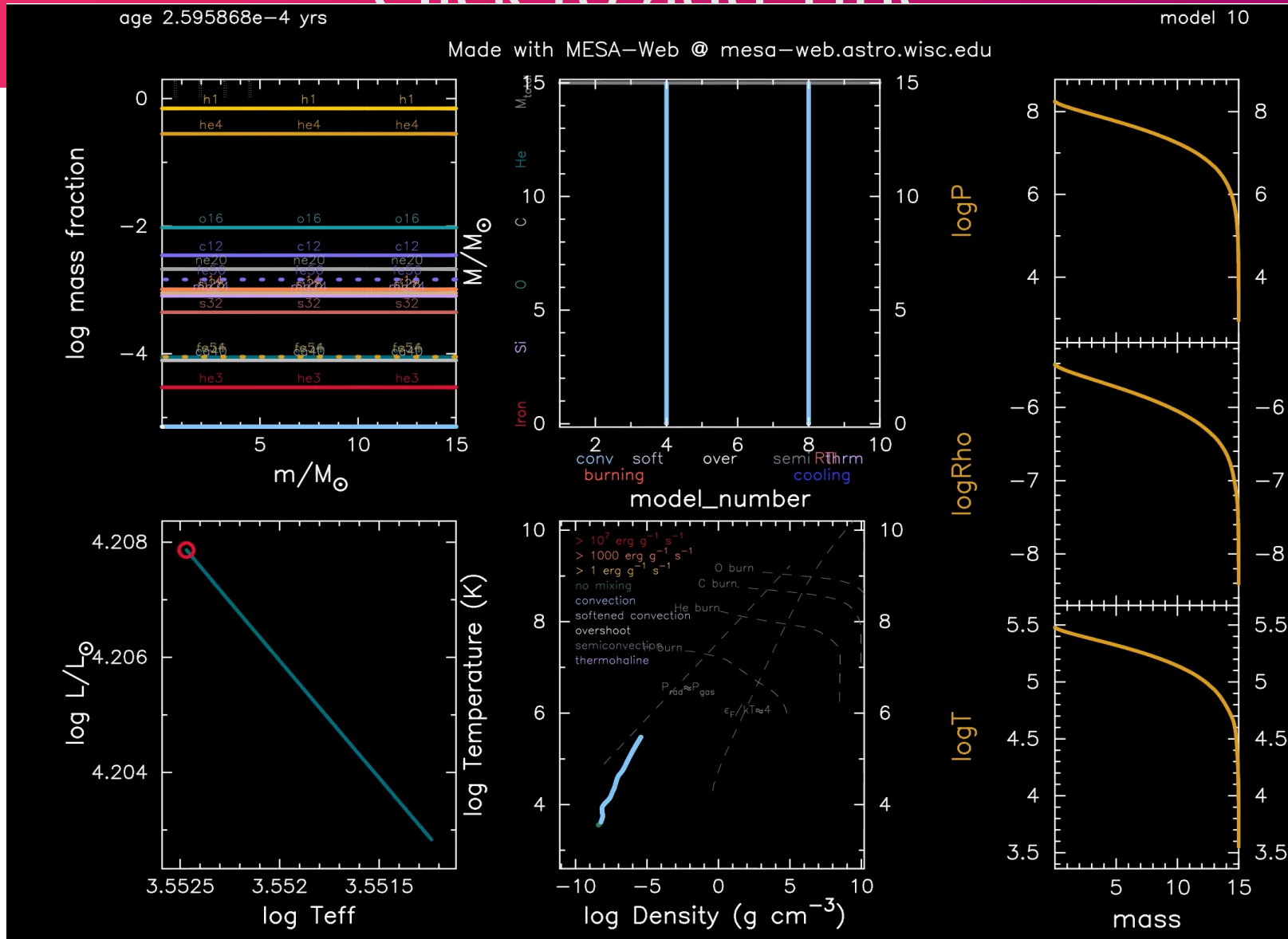
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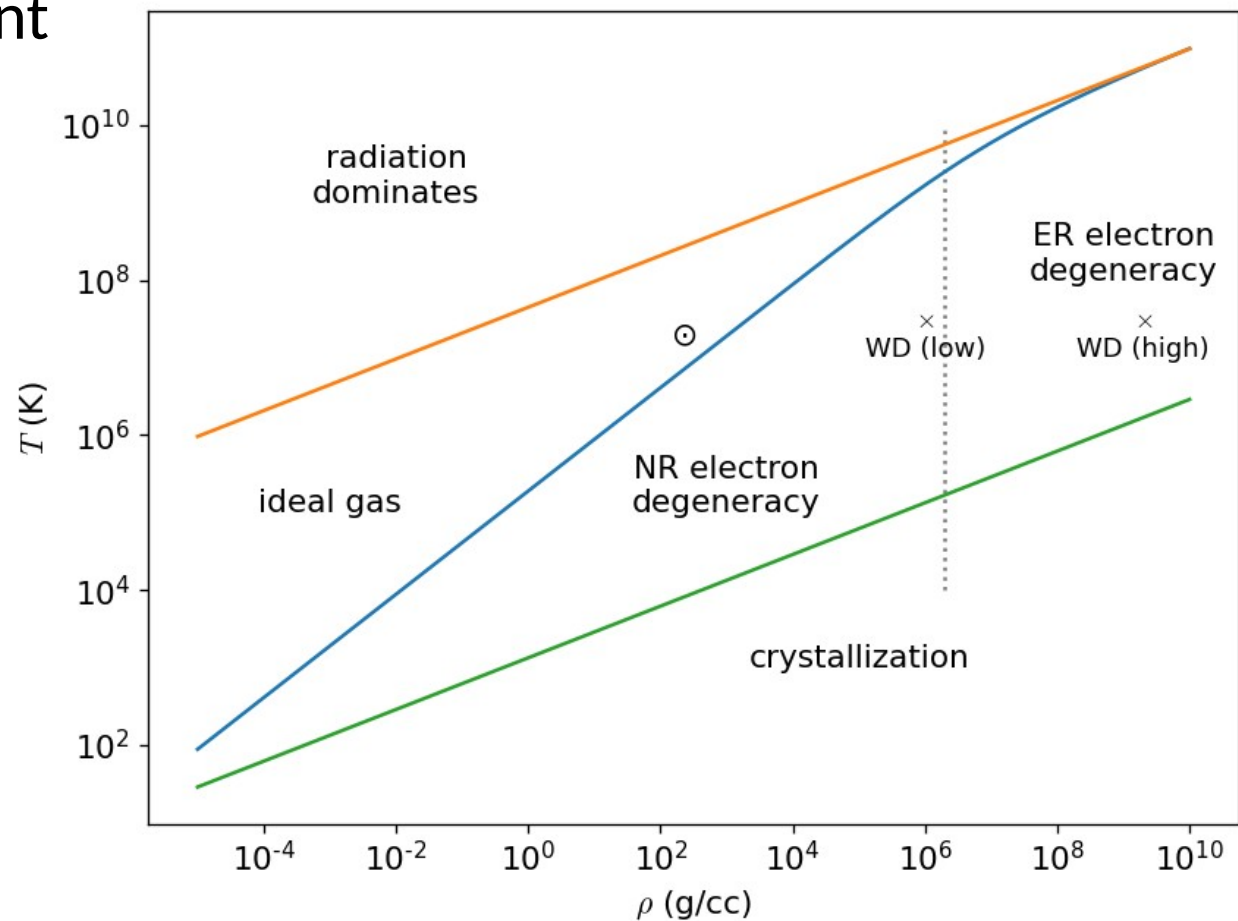


# Stellar Evolution

- T is highest at center
  - Reactions will be most energetic in center
  - Evolutionary changes are driven by exhaustion of fuel
- We can start to get a sense of stellar evolution by looking at the central conditions,  $\rho_c$ ,  $T_c$

# Log T – Log $\rho$ Plane

- In our class notebooks, we produced a diagram of when the different components of the EOS are important



# Ignition Curves

- We can also look at what fuel can burn as curves in the  $\rho$ - $T$  plane
- For simplicity, we can take

$$q = q_{\min}$$

- Most sources will ask for the energy generation to be net positive, comparing to neutrino losses

$$q = |q_{\nu}|$$

- Example:

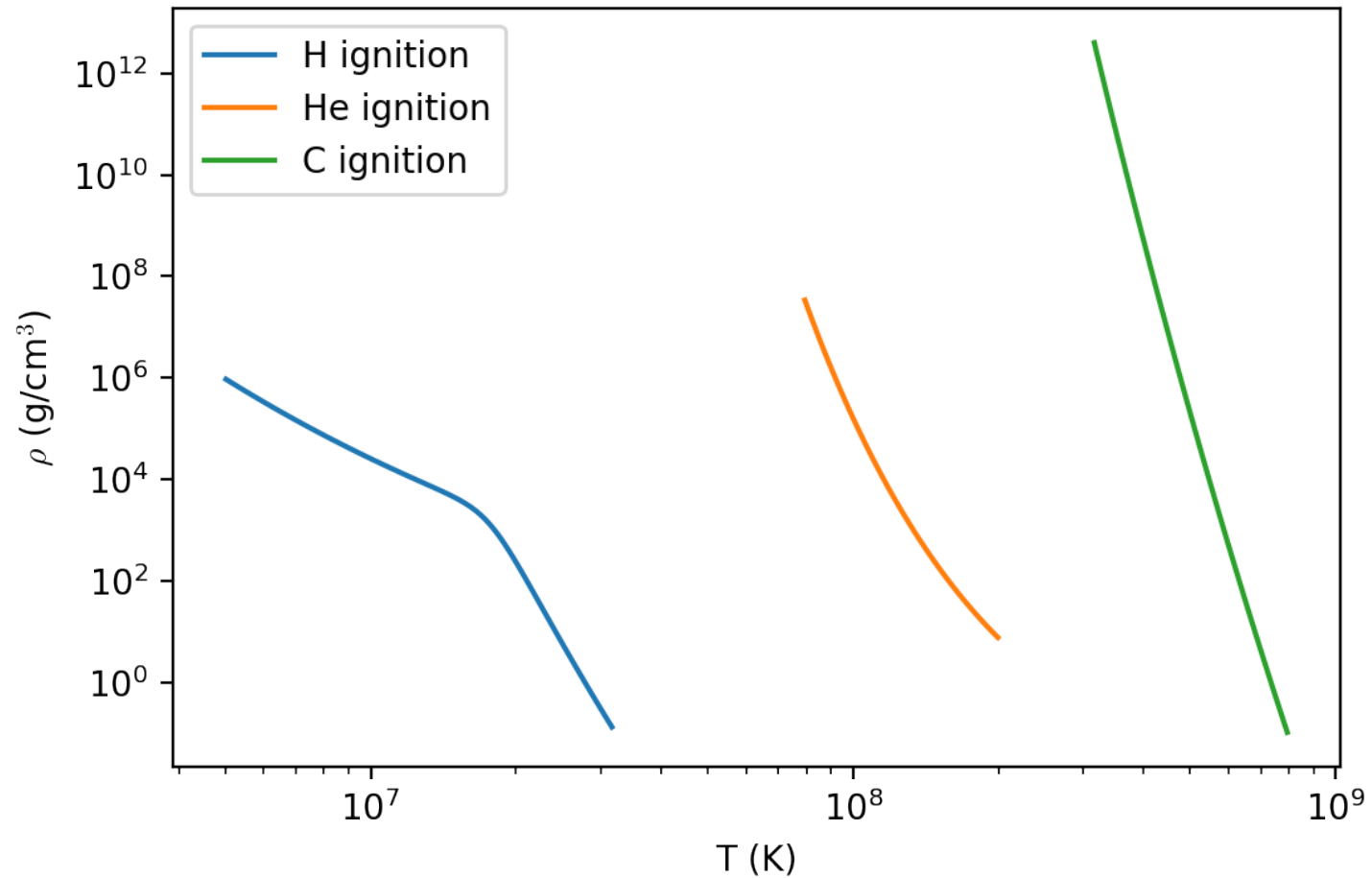
$$q_0 \rho^m T^n = q_{\min}$$

$$\log \rho = -\frac{n}{m} \log T + \frac{1}{m} \log \left( \frac{q_{\min}}{q_0} \right)$$

- Typically  $m = 1$ ,  $n \gg 1$

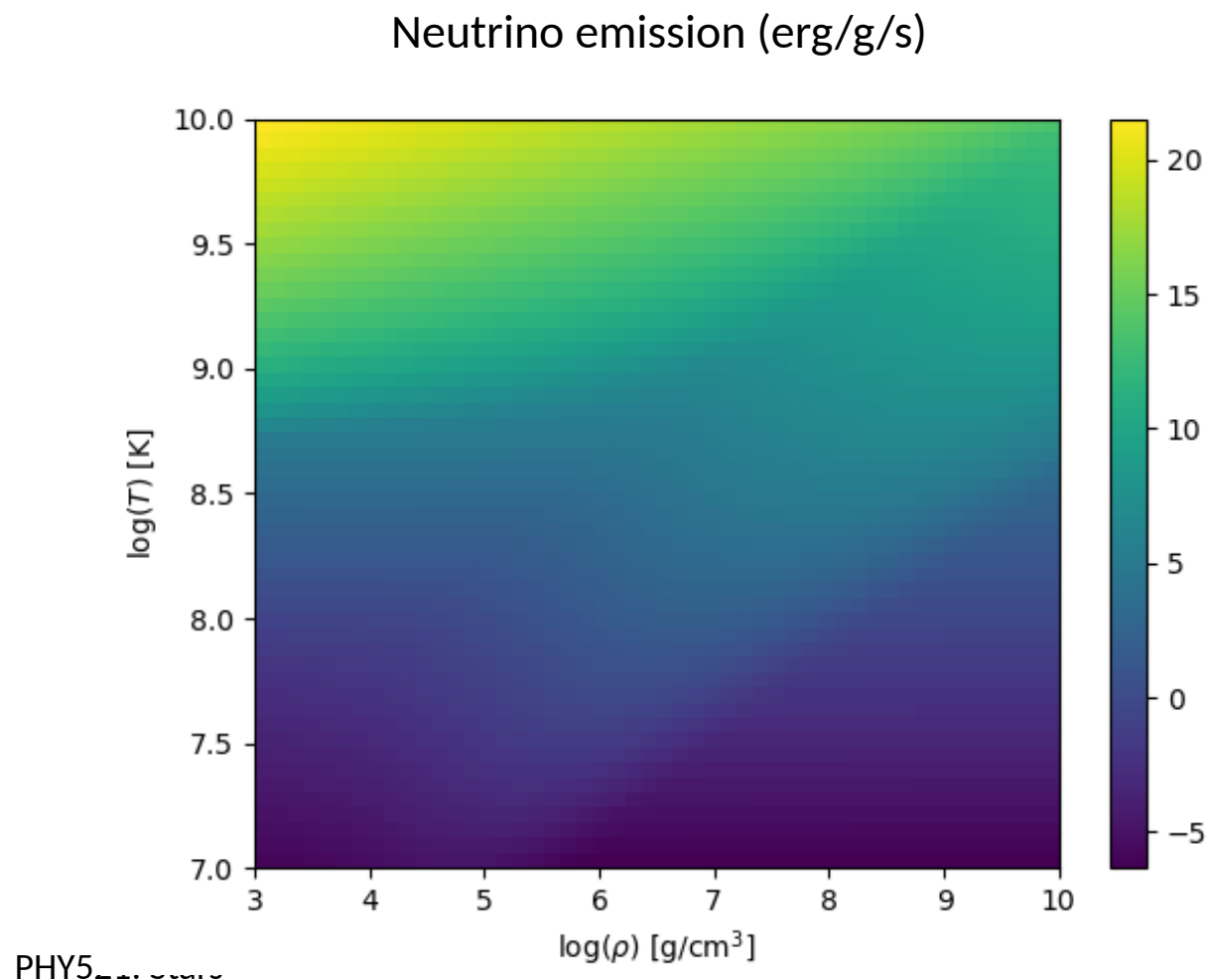


# Ignition Curves

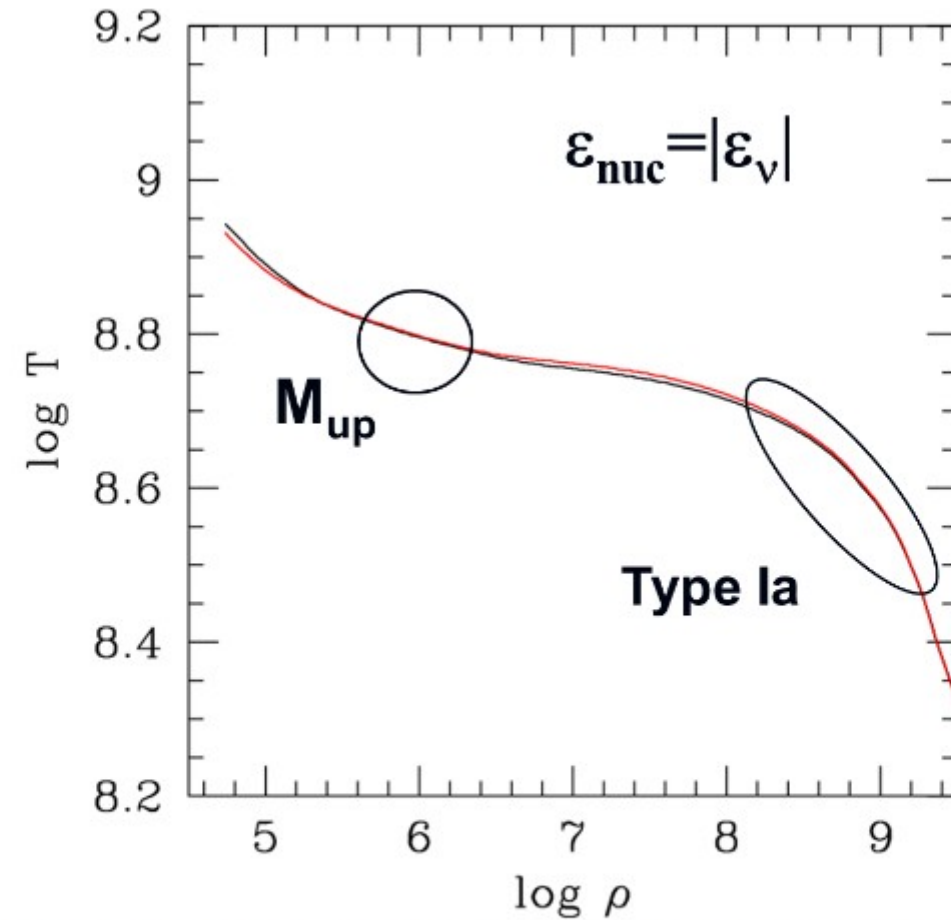


# Neutrino Losses

- Real thermal neutrino losses are much more complex:
  - Pair, photo/Bremsstrahlung, plasma neutrinos



# Ignition Curves



**Figure 3.** The carbon ignition curve (see text for the definition). The red and the black curves have been obtained under different assumptions for the neutrinos production rate, namely: [8] (red) and [9] (black). The two circled regions correspond to the typical ignition conditions for core-C burning in single massive stars and type Ia supernovae, respectively.

# Instability Zones

- We can have regions where  $\Gamma \leq 4/3$ 
  - Extremes in radiation dominated and relativistic degenerate regions
  - Iron photodisintegration (high T)
  - Pair production ( $\gamma + \gamma \rightarrow e^+ + e^-$ )
- Furthermore, nuclear burning is unstable in degenerate regions



# Evolution of Central Conditions

- Let's find a useful result from polytropes

- Start with our mass relation:

$$M_{\star} = -\frac{1}{\sqrt{4\pi}} \left( \frac{n+1}{G} \right)^{3/2} K^{3/2} \rho_c^{(3-n)/2n} \xi_1^2 \left( \frac{d\theta}{d\xi} \right)_{\xi_1}$$

- Define  $M_n \equiv -\xi_1^2 (d\theta/d\xi)_{\xi_1}$
- Solve for K:

$$K = (4\pi)^{1/2} \frac{G}{n+1} \left( \frac{M_{\star}}{M_n} \right)^{2/3} \rho_c^{-(3-n)/3n}$$

- And into the EOS:  $P_c = K \rho_c^{1+1/n}$

$$P_c = (4\pi)^{1/3} \frac{G}{n+1} \left( \frac{M_{\star}}{M_n} \right)^{2/3} \rho_c^{4/3} = (4\pi)^{1/3} B_n G M_{\star}^{2/3} \rho_c^{4/3}$$

- $B_n$  varies very slowly with  $n$ , independent of  $K$

# Evolution of Central Conditions

- How does the stellar center evolve in  $\log \rho - \log T$ ?
- From polytropes:

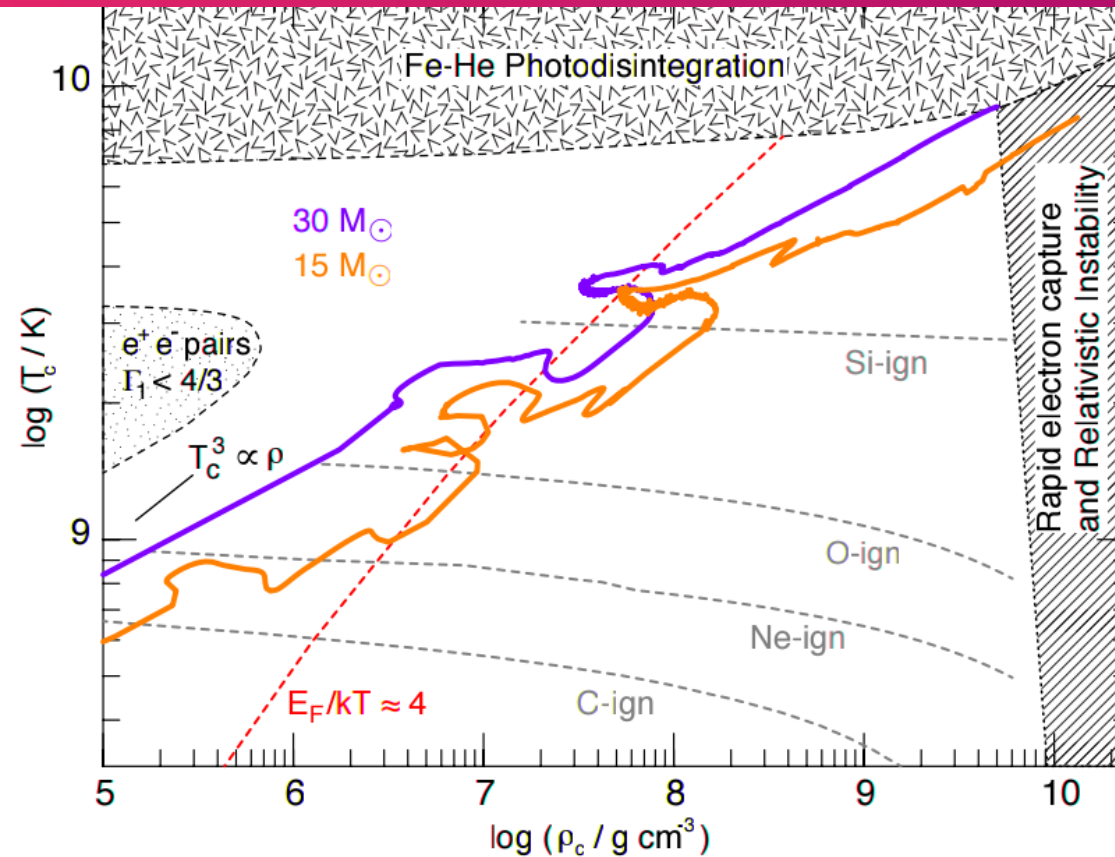
$$P_c = (4\pi)^{1/3} B_n G M^{2/3} \rho_c^{4/3}$$

- For star in ideal gas region ( $P = K_0 \rho_c T_c$ ):

$$\rho_c = \frac{K_0^3}{4\pi B_n^3 G^3} \frac{T_c^3}{M^2}$$

- These are parallel lines for different masses
- For same  $\rho_c$ , higher mass stars have higher  $T_c$

# Evolution of Central Conditions



**Figure 24.** Evolution of  $T_c$  and  $\rho_c$  in solar metallicity, non-rotating  $M_i = 15$  and  $30 M_\odot$  pre-supernova models. The curves are calculated using an in-situ 204 isotope reaction network. Locations of the core carbon, neon, oxygen, and silicon ignition are labeled, as is the scaling relation  $T_c^3 \propto \rho_c$ , and the  $E_F/k_B T \approx 4$  electron degeneracy curve. Regions dominated by electron-positron pairs, photodisintegration, and rapid electron capture are shaded and labeled.

(Paxton et al. MESA III paper)

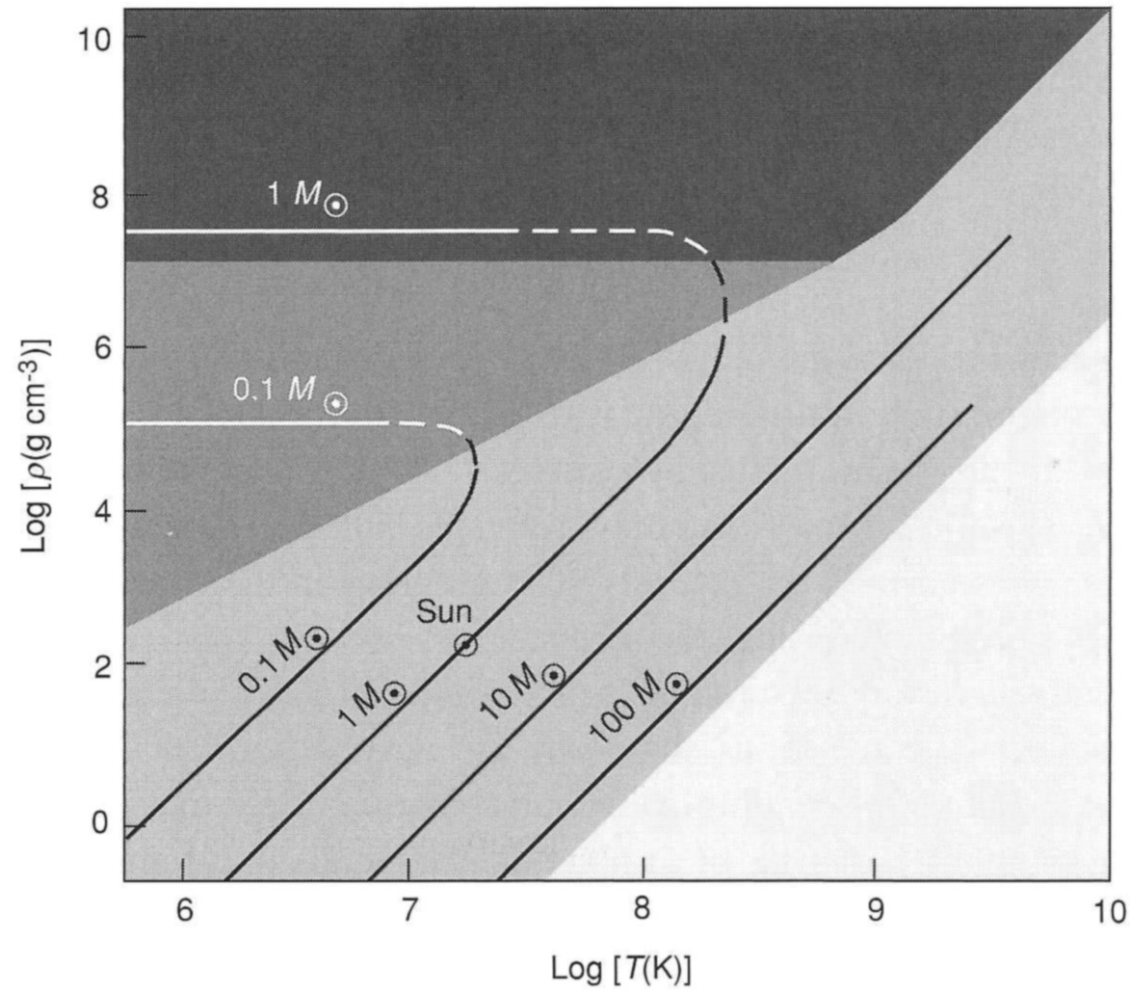
# Evolution of Central Conditions

- For star in degenerate region,  $P = K_1 \rho^{5/3}$ :

$$\rho_c = 4\pi \left( \frac{B_n G}{K_1} \right)^3 M^2$$

- No T dependence, only depends on M
- Maximum mass support by electrons is Chandrasekhar mass
  - Stars with  $M > M_{\text{Ch}}$  will not enter this region

# Evolution of Central Conditions



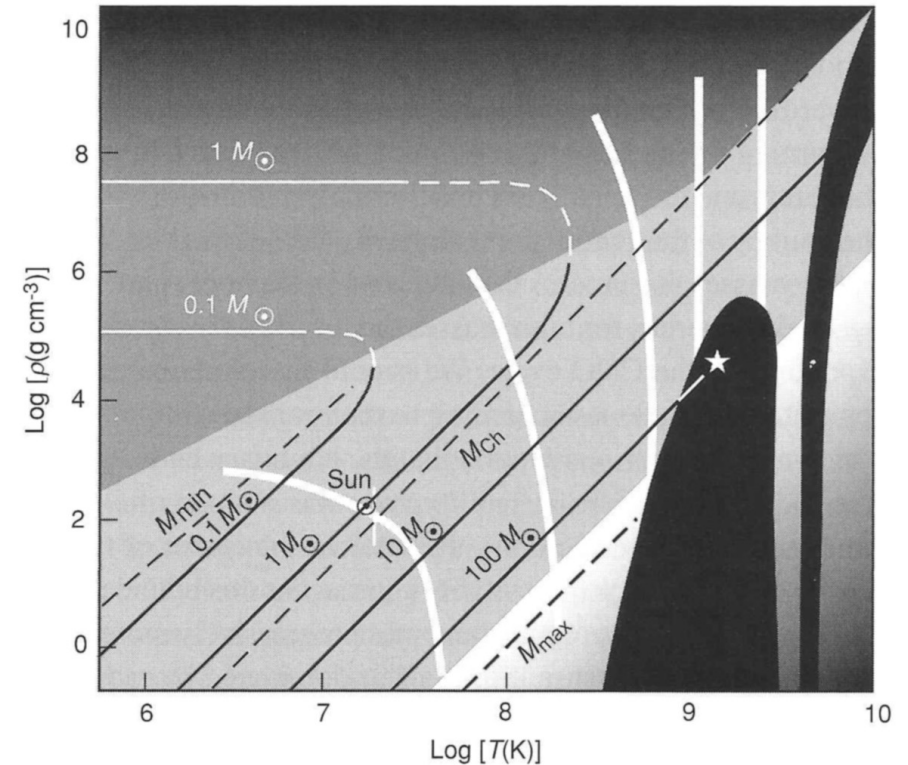
(Prialnik)

**Figure 7.4** Relation of central density to central temperature for stars of different masses within the stable ideal gas and degenerate gas zones.



# Overview of Evolution

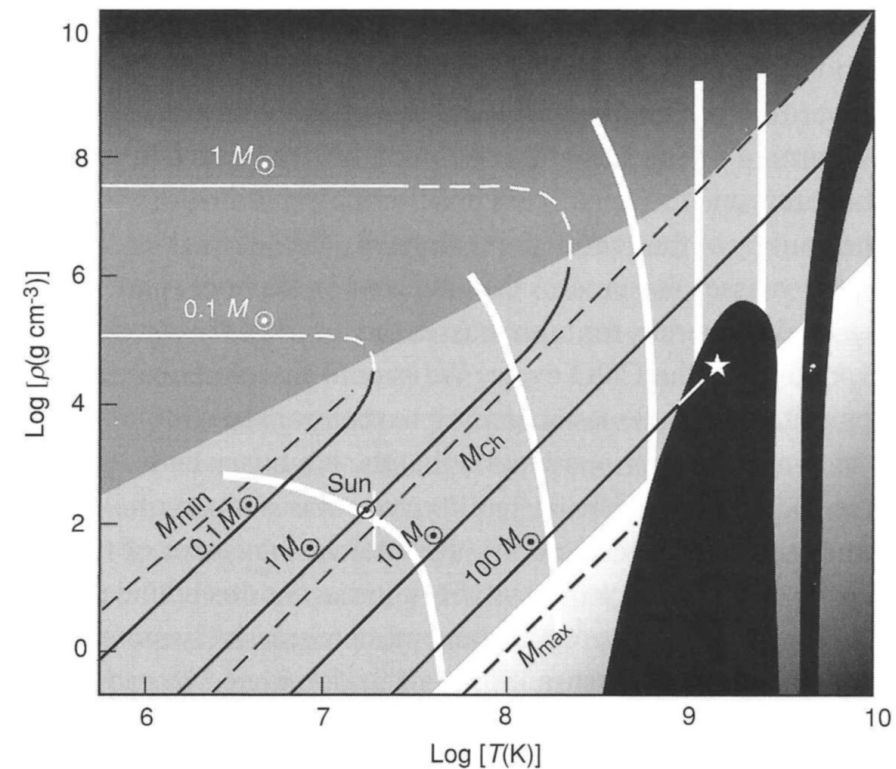
- Star forms at low  $p$ - $T$  (gas cloud)
  - Start at lower left corner
  - No burning: star radiates gravitational potential energy and contracts / heats
  - Star follows  $\rho_c \propto T_c^3$
- Star crosses H ignition line
  - Core H burning begins
  - Thermal equilibrium established
  - We stay here a long time: main sequence
  - Where we cross H line determines pp or CNO



**Figure 7.5** Schematic illustration of the evolution of stars according to their central temperature-density tracks.

# Overview of Evolution

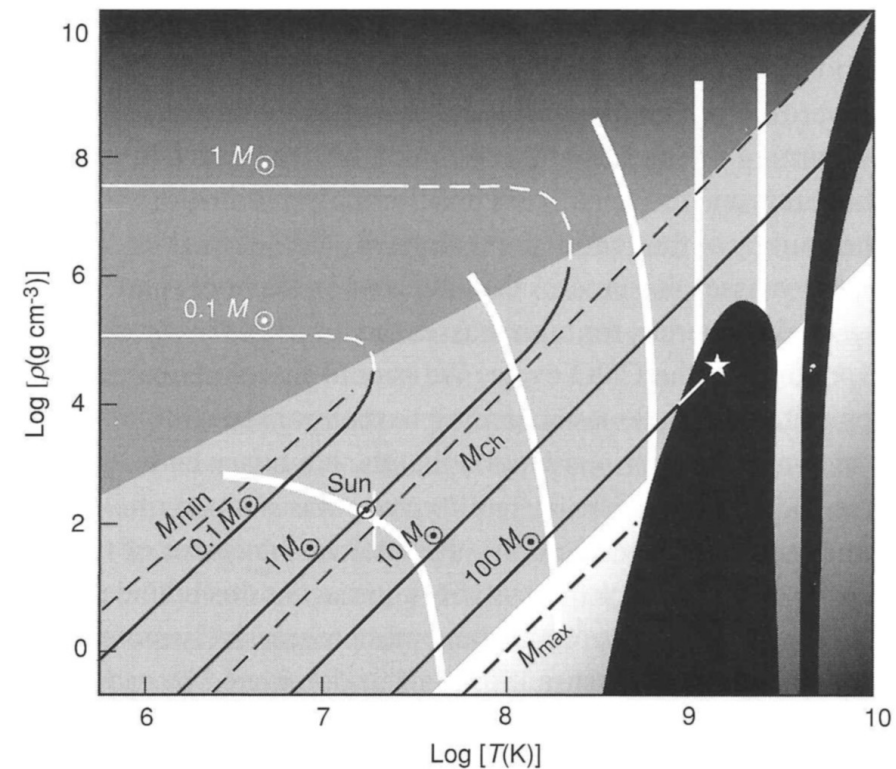
- If we are in the radiation zone, then we are unstable
  - Most massive stars will be here and not form
  - Upper limit to stellar masses
- Lower limit to stellar masses is where H curve stops ( $T \sim 10^6$  K)
  - Somewhere near  $M \sim 0.1 M_{\odot}$



**Figure 7.5** Schematic illustration of the evolution of stars according to their central temperature-density tracks.

# Overview of Evolution

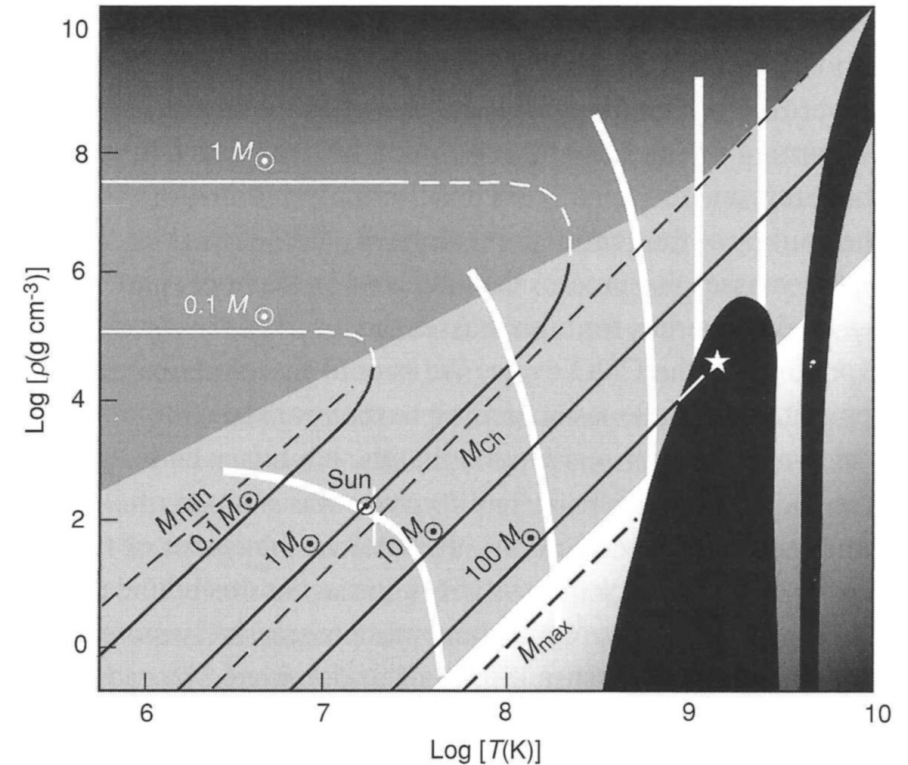
- H core exhaustion: core contracts and heats up
  - Climb up the curve
  - Low mass stars cross into degeneracy
  - Higher masses cross He ignition line
    - Lowest masses here cross close to degeneracy: He flash



**Figure 7.5** Schematic illustration of the evolution of stars according to their central temperature-density tracks.

# Overview of Evolution

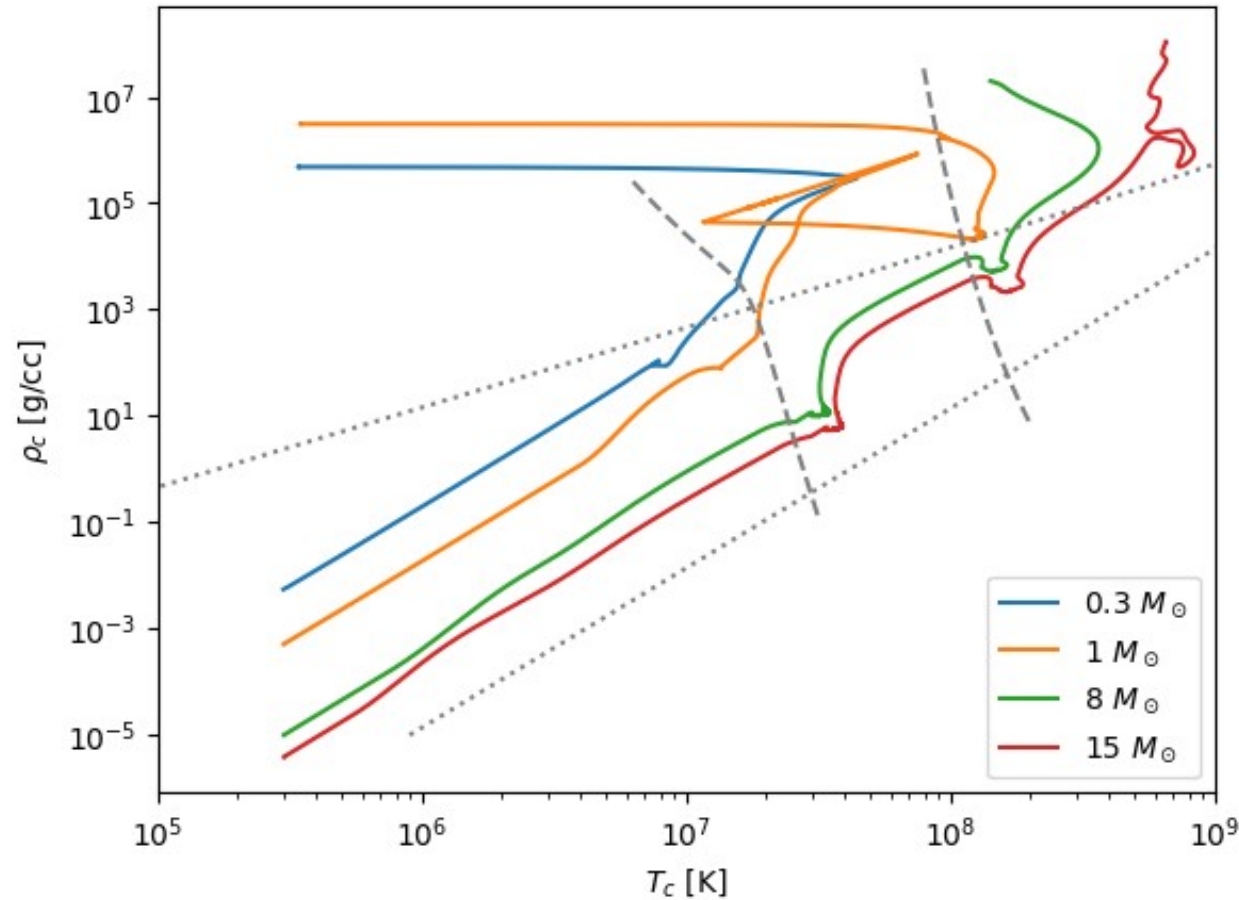
- Process continues
  - Higher masses (core mass > Chandrasekhar mass) continue to reach other burning zones
  - Evolution passes shortly at each new fuel
- When we reach Fe, photodisintegration kicks in: instability
- Highest masses have path enter pair-instability region—instability before Fe burning



**Figure 7.5** Schematic illustration of the evolution of stars according to their central temperature-density tracks.

# Visualizing Evolution

- Let's compare this picture to the MESA models:  
[https://zingale.github.io/stars/notebooks/stellar\\_evolution/stellar-models.html](https://zingale.github.io/stars/notebooks/stellar_evolution/stellar-models.html)



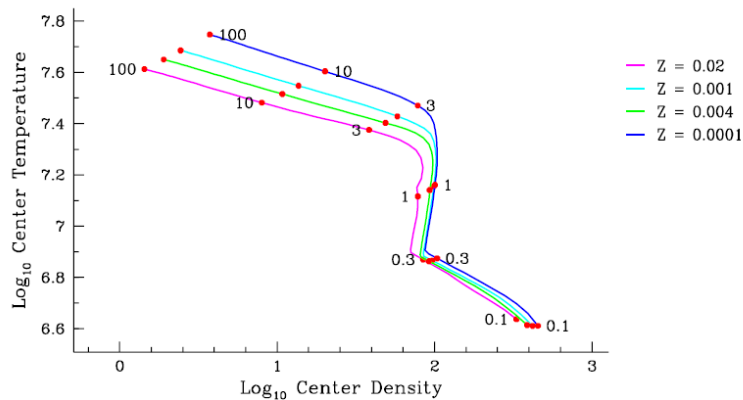
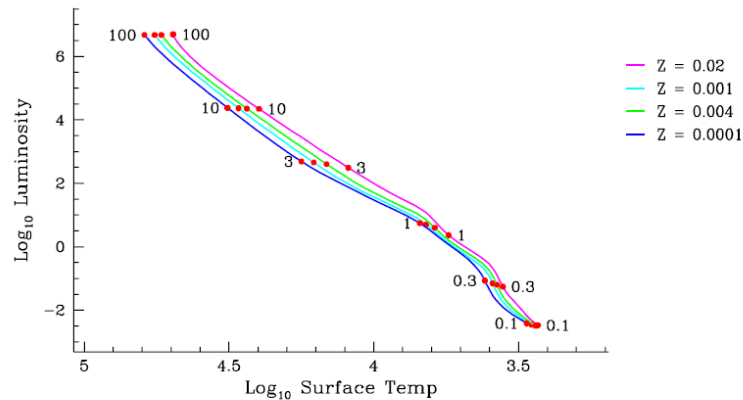
# ZAMS

- If we redid dimensional analysis from week 2 but with Kramer's opacity:

$$L \sim M^{71/13} Z^{-14/13}$$

$$L \sim T_{\text{eff}}^{284/69} Z^{8/23}$$

- Notice that at the low mass end:
  - Stars with the same mass, but lower  $Z$  are more luminous
  - Lower  $Z$  have the main sequence beneath that of higher  $Z$

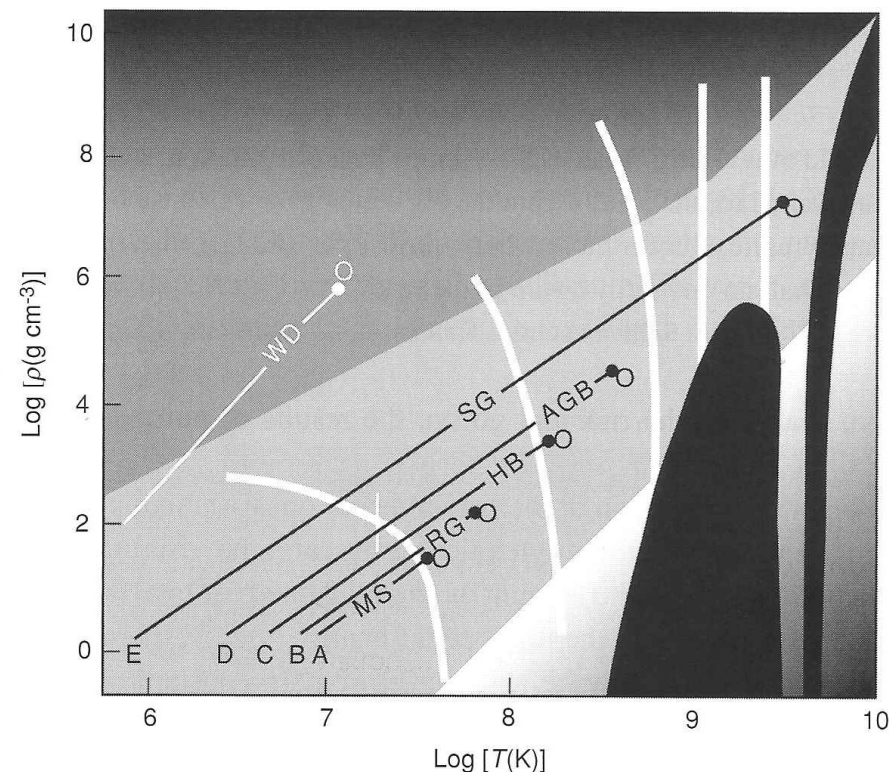


ZAMS Hertzsprung-Russell and Temperature-Density Plots. Numbers show mass in solar units.



# Late Stellar Evolution

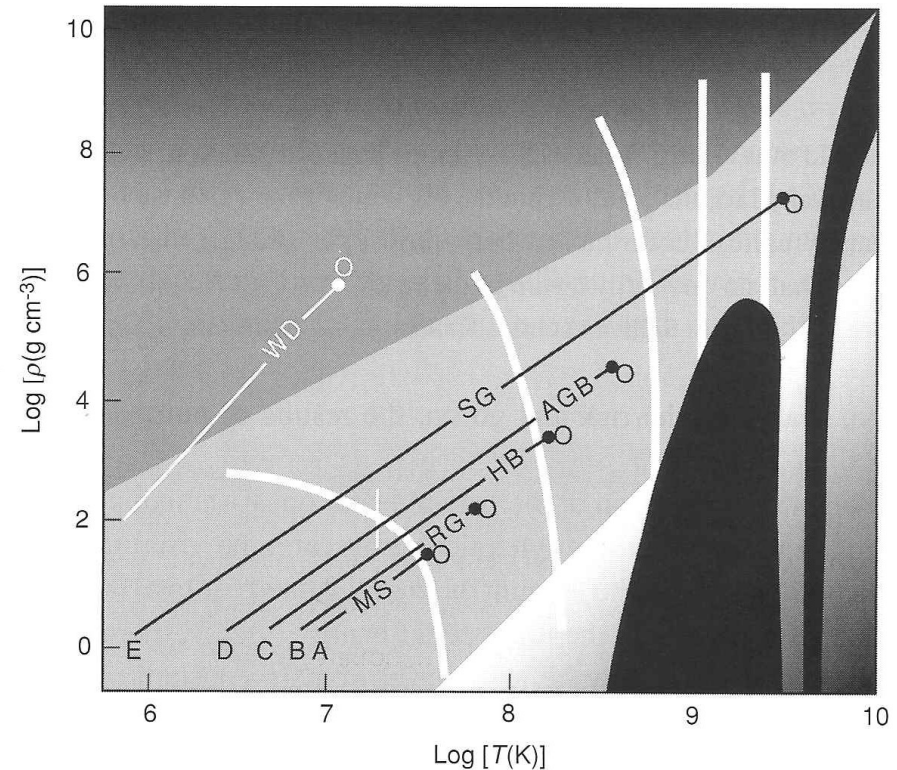
- We can use the same  $\log \rho - \log T$  plane to look at the structure of a star at some instant in time
  - Now the curves show  $(\rho(m), T(m))$
- Ex: 10 solar mass star
- A is main sequence: core H burning
  - Core is on the H ignition curve



**Figure 7.6** Schematic illustration of the stellar configuration in different evolutionary phases for a  $10M_{\odot}$  star (A, B, C, D, E) and a white dwarf (WD).

# Late Stellar Evolution

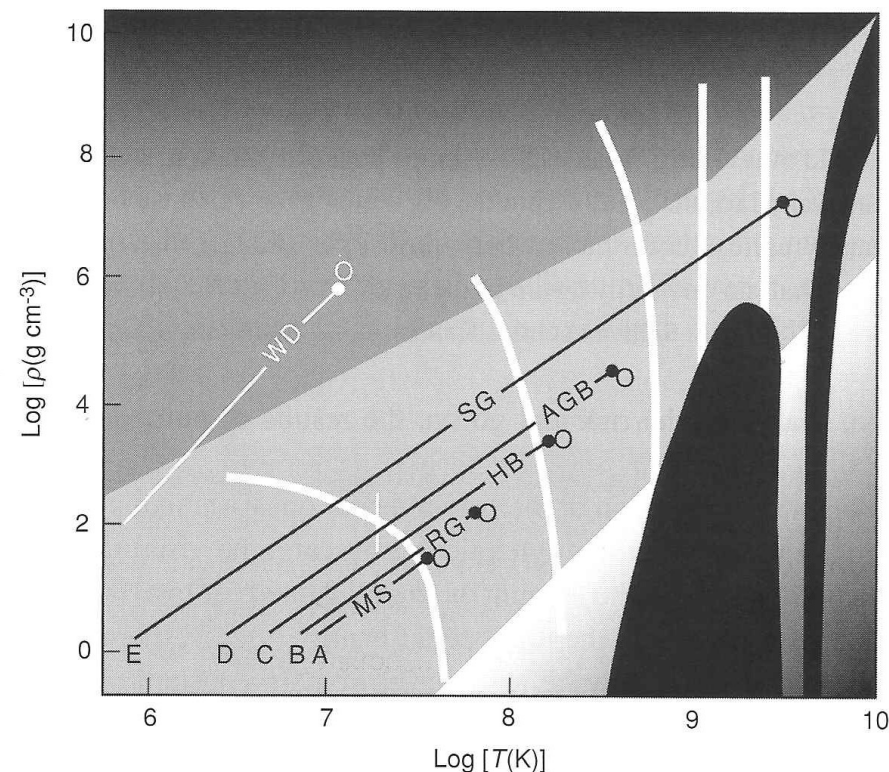
- B: core H depleted
  - Core contracts
  - Other mass elements cross the H ignition curve: shell burning
  - Contraction is quasi-static
    - Virial theorem applies
    - Assume thermal equilibrium: both  $U$  and  $\Omega$  conserved independently
- Core contraction:  $\Omega$  increases in magnitude at core  $\rightarrow$  outer layers expand
- Core heating means  $U$  increases at core, so  $U$  must decrease in envelope: red giant



**Figure 7.6** Schematic illustration of the stellar configuration in different evolutionary phases for a  $10M_{\odot}$  star (A, B, C, D, E) and a white dwarf (WD).

# Late Stellar Evolution

- C: He ignition reached
  - Now 2 ignition curves are crossed: He core burning and H shell burning
- D: He core exhaustion, so core contracts again
  - Two shell burning phases
- E: multiple burning shells, no more core fuel... lead up to core collapse



**Figure 7.6** Schematic illustration of the stellar configuration in different evolutionary phases for a  $10 M_{\odot}$  star (A, B, C, D, E) and a white dwarf (WD).

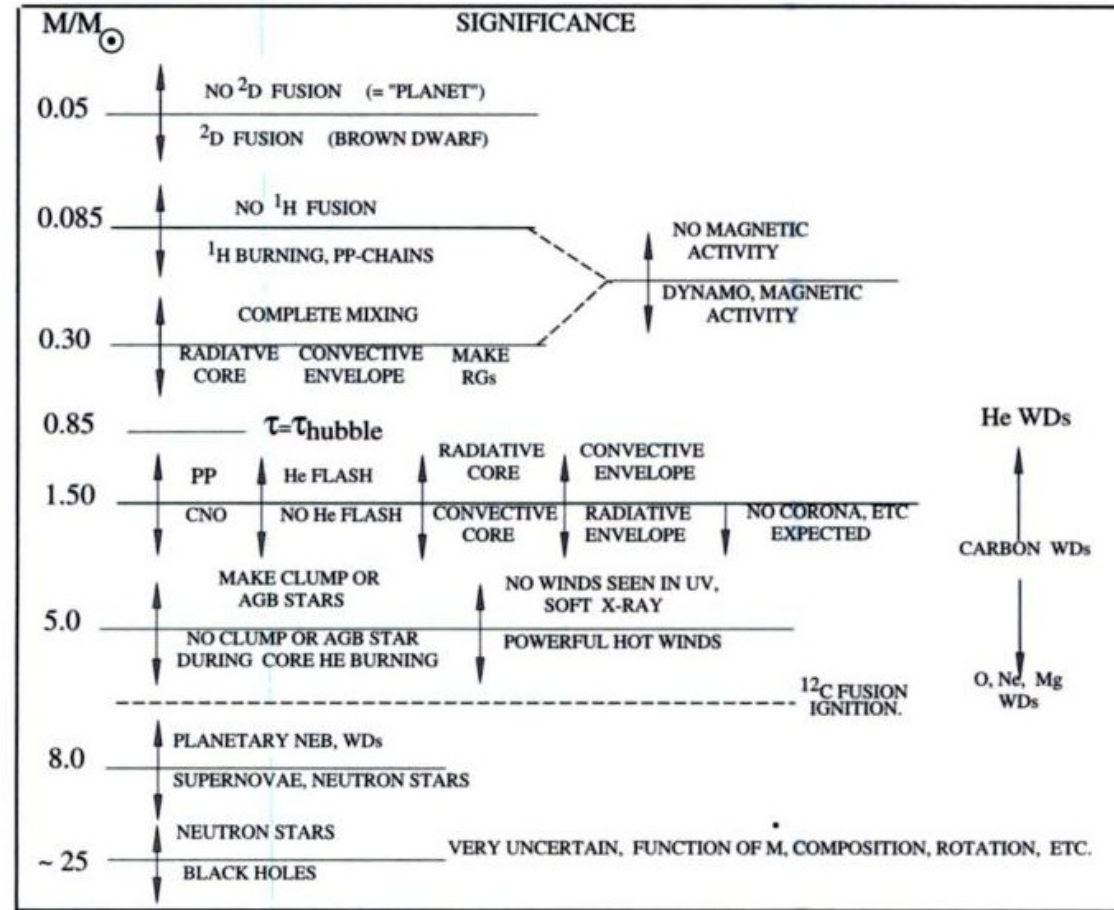
# Massive Star Evolution

- Massive stars ( $M > 8 M_{\odot}$ ) ignite He and C under non-degenerate conditions
  - Stars above  $11 M_{\odot}$  also ignite heavier fuels beyond carbon non-degenerately
  - Some uncertainty due to mass loss
- Note: this simple picture did not include mass loss by winds



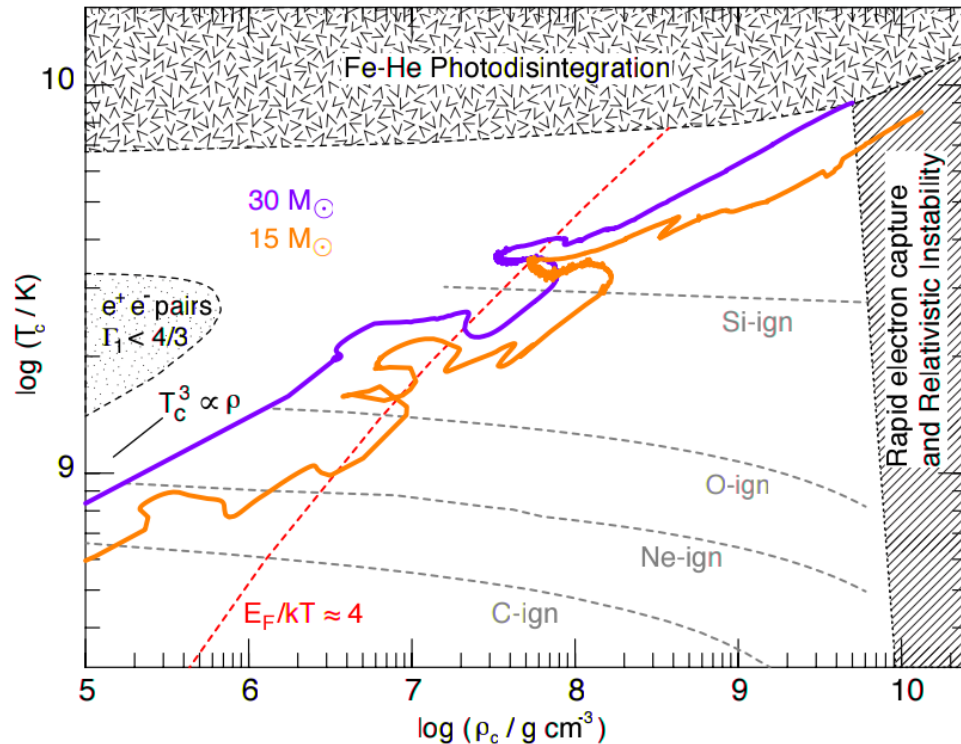
# Mass Cuts

- Mass plays a big role in the outcome of stellar evolution



**Fig. 2.4.** Our "Mass Cut" diagram showing the fate of single stars in various mass classes. See text. (Hansen, Kawaler, Trimble)

# High Mass Evolution



**Figure 24.** Evolution of  $T_c$  and  $\rho_c$  in solar metallicity, non-rotating  $M_i = 15$  and  $30 M_\odot$  pre-supernova models. The curves are calculated using an in-situ 204 isotope reaction network. Locations of the core carbon, neon, oxygen, and silicon ignition are labeled, as is the scaling relation  $T_c^3 \propto \rho_c$ , and the  $E_F/k_B T \approx 4$  electron degeneracy curve. Regions dominated by electron-positron pairs, photodisintegration, and rapid electron capture are shaded and labeled.

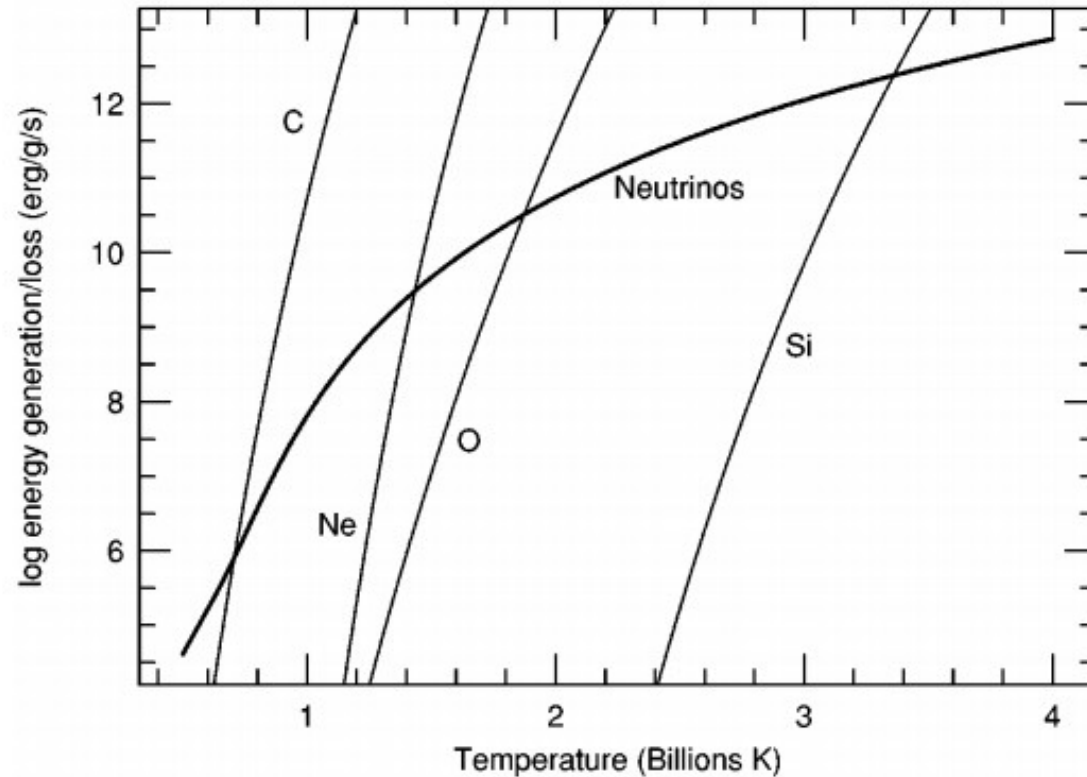
- Core evolution follows (roughly),  $T^3 \sim \rho$ 
  - This is also what Eddington standard model shows
- Neutrino losses become important
  - At  $T > 5 \times 10^8$  K, this dominates energy losses
  - Burning ignites where  $\epsilon_v \sim \epsilon_{\text{nuc}}$
- Burning from C to Fe core is so fast that the outer layers don't really have time to adjust
  - We'll be mostly stationary on the HR diagram

(Paxton et al. MESA III paper)



# Burning Lifetimes

- Lifetimes can be estimated using the energy generate rate where  $\epsilon_v \sim \epsilon_{\text{nuc}}$



**Figure 12.6.** Energy generation rate and neutrino loss rate during the advanced evolution of a massive star. The stellar center is assumed to follow a track approximating that shown in Fig. 12.5. The intersections of the nuclear burning lines with the neutrino loss line define the burning temperature of the corresponding fuel. Figure from Woosley, Heger & Weaver (2002).

(via Onno Pols notes)

# Mass Loss

- Mass loss is very uncertain
- In 1D, various prescriptions are used—see Renzo et al. 2017 for a review
  - Parameterized mass loss rate averaged over timestep
  - $\dot{M} = \dot{M}(L, T_{\text{eff}}, Z, \dots)$
  - Efficiency factor,  $\eta$ , accounts for uncertainties
  - Different mass loss algorithms for each phase of evolution
- Mass loss for high-mass stars has 3 phases:
  - Hot:  $T_{\text{eff}} > 15000 \text{ K}$
  - Cool:  $T_{\text{eff}} < 15000 \text{ K}$
  - Wolf-Rayet: surface H mass fraction,  $X_s < 0.4$

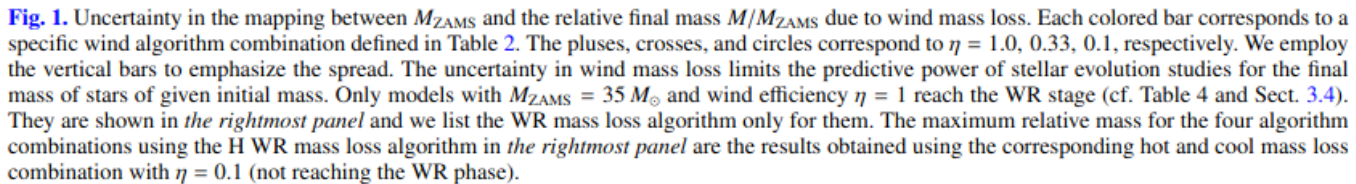
**Table A.1.** Functional dependence of the mass loss rate  $\dot{M}$  on stellar and wind parameters for each algorithm combination and mass loss phase.**Table 1.** Approximate scaling of the mass loss rate with effective temperature,  $\log_{10}(\dot{M}) \propto \alpha \log_{10}(T_{\text{eff}})$ , by the considered mass loss algorithms.

|      | ID                           |
|------|------------------------------|
| Hot  | V                            |
|      | K                            |
| Cool | dJ                           |
|      | NJ                           |
|      | vL                           |
| WR   | NL                           |
|      | H ( $L > 4.5 L_{\odot}$ )    |
|      | H ( $L \leq 4.5 L_{\odot}$ ) |

**Notes.** Appendix A provides the full function implemented in MESA and Table A.1 lists quantities.

| ID   | V-dJ-NL                                                                                                                                                                                     | V-NJ-NL                      | V-vL-NL                          | V-dJ-H                                                     | V-NJ-H                       | V-vL-H                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------|------------------------------------------------------------|------------------------------|----------------------------------|
| Hot  | $L^{2.210} M^{-1.339} \left(\frac{v_{\infty}}{2v_{\text{esc}}}\right)^{-1.601} T_{\text{eff}}^{1.07} Z^{0.85}$ if $T_{\text{eff}} < 22.5$ kK                                                |                              |                                  |                                                            |                              |                                  |
|      | $L^{2.194} M^{-1.313} \left(\frac{v_{\infty}}{2v_{\text{esc}}}\right)^{-1.226} Z^{0.85} T_{\text{eff}}^{(0.933-10.92 \log_{10}(T_{\text{eff}}/40\text{kK}))}$ if $T_{\text{eff}} > 27.5$ kK |                              |                                  |                                                            |                              |                                  |
| Cool | $L^{1.769} T_{\text{eff}}^{-1.676}$                                                                                                                                                         | $L^{1.24} M^{0.16} R^{0.81}$ | $L^{1.05} T_{\text{eff}}^{-6.3}$ | $L^{1.769} T_{\text{eff}}^{-1.676}$                        | $L^{1.24} M^{0.16} R^{0.81}$ | $L^{1.05} T_{\text{eff}}^{-6.3}$ |
| WR   | $L^{1.29} Y_s^{1.73} Z_s^{0.47}$                                                                                                                                                            |                              |                                  | $L^{1.5} 10^{-2.85 X_s}$ if $\log_{10}(L/L_{\odot}) > 4.5$ |                              |                                  |
|      |                                                                                                                                                                                             |                              |                                  | $L^{6.8}$ if $\log_{10}(L/L_{\odot}) \leq 4.5$             |                              |                                  |
| ID   | K-dJ-NL                                                                                                                                                                                     | K-NJ-NL                      | K-vL-NL                          | K-dJ-H                                                     | K-NJ-H                       | K-vL-H                           |
| Hot  | $L^{1.779} v_{\text{th}}^{-1.169} [GM(1 - \Gamma_E)]^{0.610}$                                                                                                                               |                              |                                  |                                                            |                              |                                  |
| Cool | $L^{1.769} T_{\text{eff}}^{-1.676}$                                                                                                                                                         | $L^{1.24} M^{0.16} R^{0.81}$ | $L^{1.05} T_{\text{eff}}^{-6.3}$ | $L^{1.769} T_{\text{eff}}^{-1.676}$                        | $L^{1.24} M^{0.16} R^{0.81}$ | $L^{1.05} T_{\text{eff}}^{-6.3}$ |
| WR   | $L^{1.29} Y_s^{1.73} Z_s^{0.47}$                                                                                                                                                            |                              |                                  | $L^{1.5} 10^{-2.85 X_s}$ if $\log_{10}(L/L_{\odot}) > 4.5$ |                              |                                  |
|      |                                                                                                                                                                                             |                              |                                  | $L^{6.8}$ if $\log_{10}(L/L_{\odot}) \leq 4.5$             |                              |                                  |

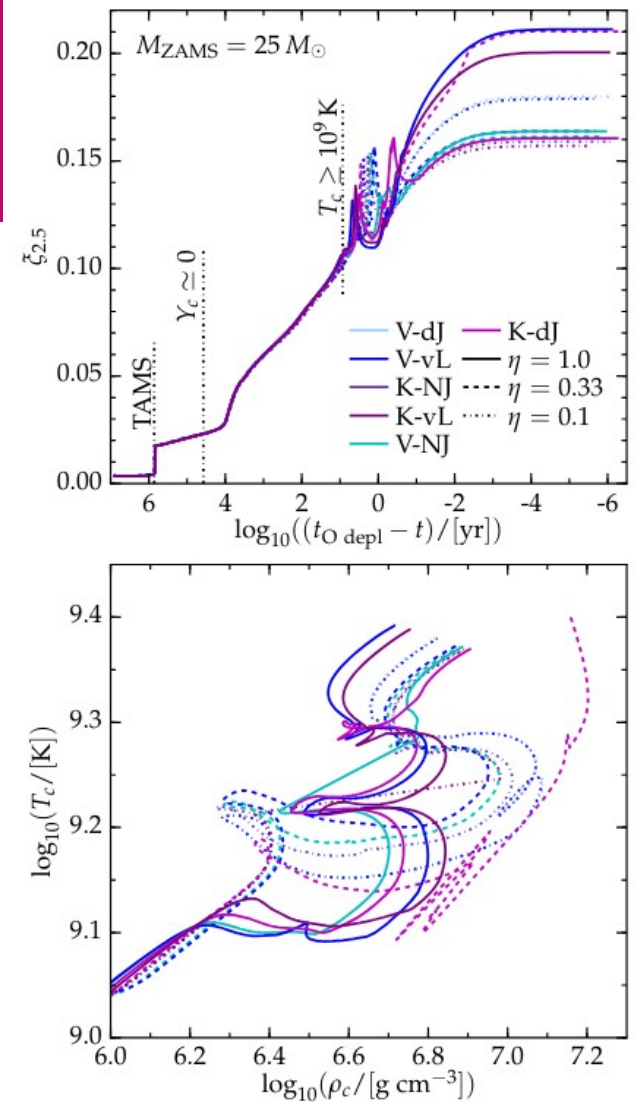
**Notes.**  $L$  is the luminosity,  $M$  is the mass,  $v_{\text{esc}}$  is the escape velocity,  $v_{\infty}$  is the final velocity of the wind,  $v_{\text{th}}$  is the thermal velocity,  $T_{\text{eff}}$  is the effective temperature,  $R$  the radius,  $\Gamma_E = L/L_{\text{Edd}}$  the Eddington ratio, and  $X_s$ ,  $Y_s$ , and  $Z_s$  are the surface hydrogen abundance, helium abundance, and metallicity, respectively. The scalings are obtained from the algorithms described in Appendix A, and errors and overall multiplying factors are omitted for the sake of brevity. See Table 2 for the naming convention for mass loss algorithm combinations.



# Mass Loss

- Different mass loss prescriptions affect the core evolution
  - Compactness parameter:

$$\xi_M \equiv \frac{M/M_\odot}{R(M)/1000 \text{ km}}$$



**Fig. 4.** *Top panel:* time evolution of the compactness parameter  $\xi_{2.5}$  for all  $M_{\text{ZAMS}} = 25 M_\odot$  models computed until oxygen depletion. *Bottom panel:* central temperature and density for the same models from when the central density increases above  $10^6 \text{ g cm}^{-3}$ . All curves end at oxygen depletion ( $X_c(^{16}\text{O}) \leq 0.04$  and  $X_c(^{28}\text{Si}) \geq 0.01$ ). The color indicates the wind algorithm combination, labeled according to Table 2. Dot-dashed, dashed, and solid curves are calculated using  $\eta = 0.1, 0.33, 1.0$ , respectively. We only show models listed in Table 7. The vertical dot-dashed lines in the top panel approximately indicate core hydrogen exhaustion (TAMS), core helium exhaustion ( $Y_c \approx 0$ ), and end of the mass loss phase ( $T_c \geq 10^9 \text{ K}$ ).

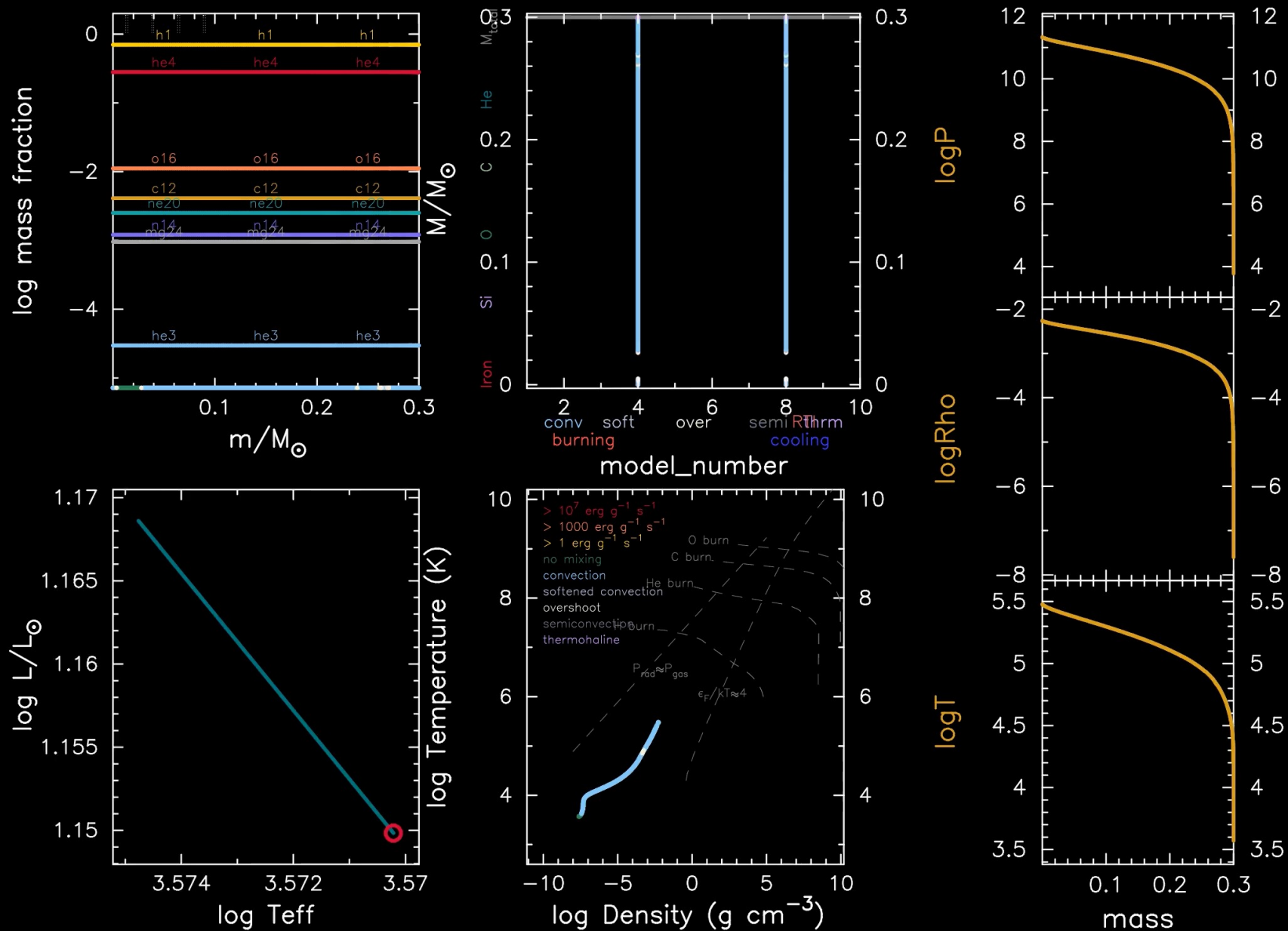


# Looking Again At Stellar Evolution

- Let's look at results of stellar evolution calculations again, now that we've learned about the physics that goes into the models

model 10

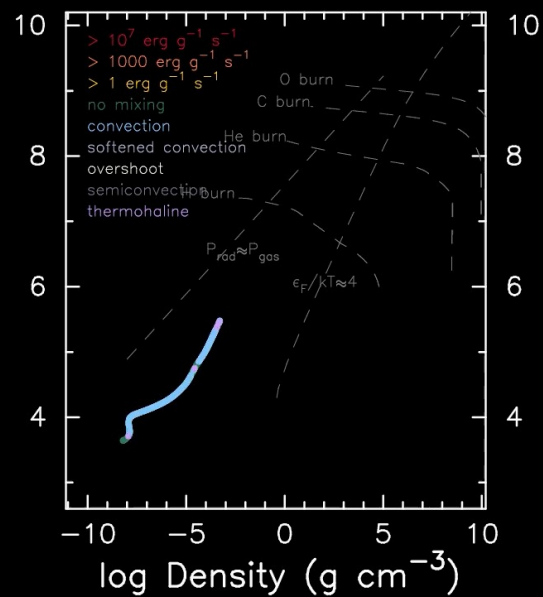
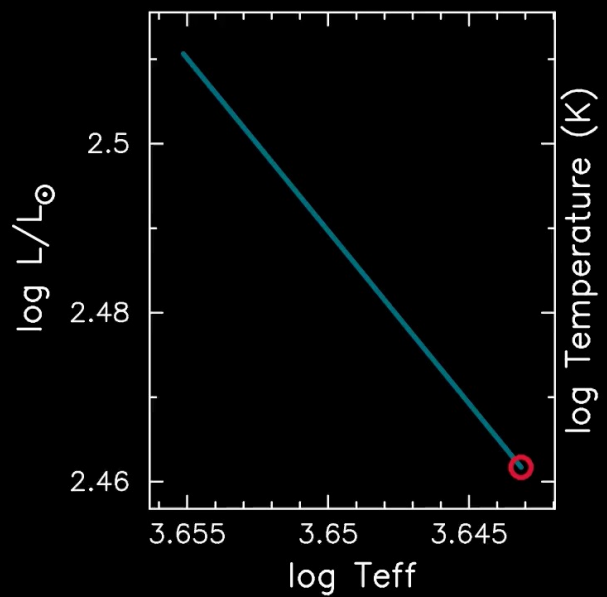
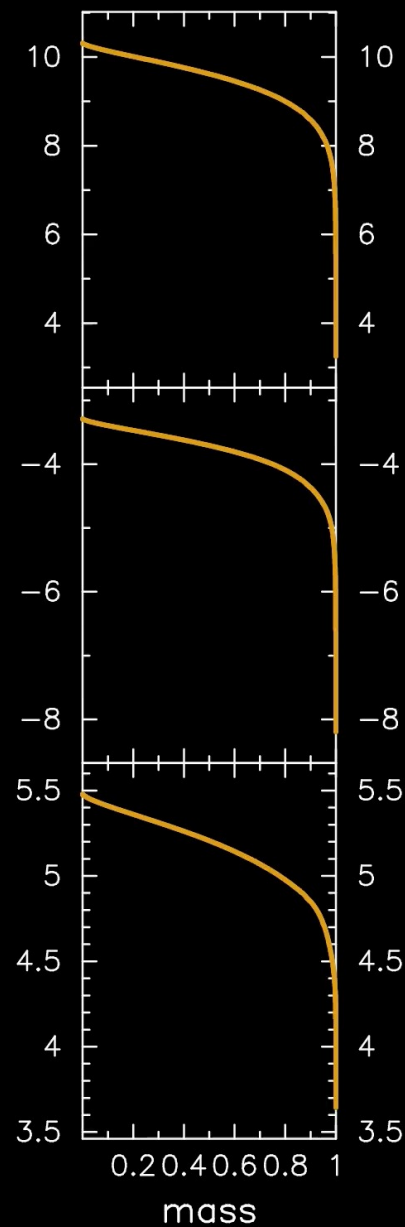
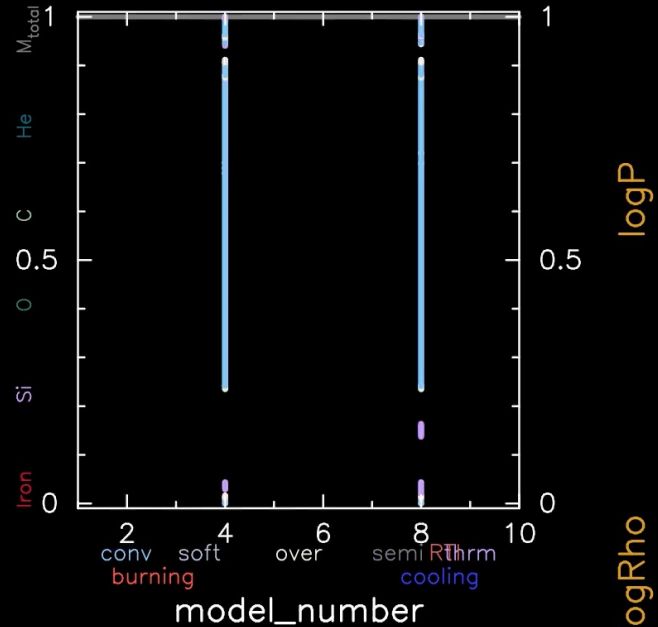
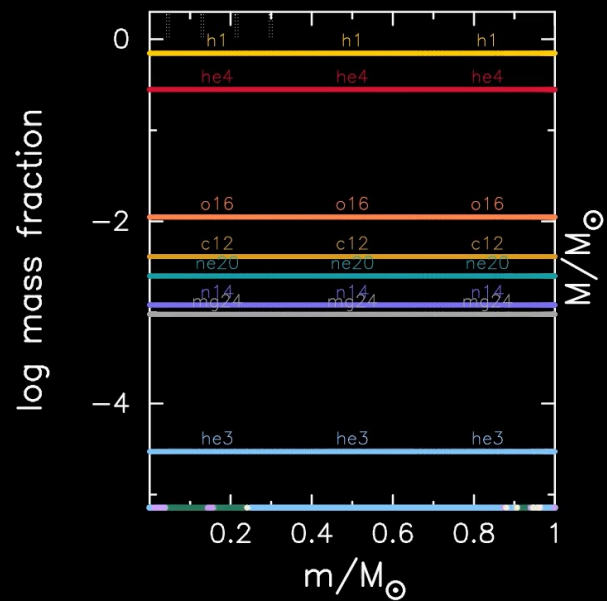
Made with MESA—Web @ [mesa-web.astro.wisc.edu](http://mesa-web.astro.wisc.edu)



age 2.595868e-4 yrs

model 10

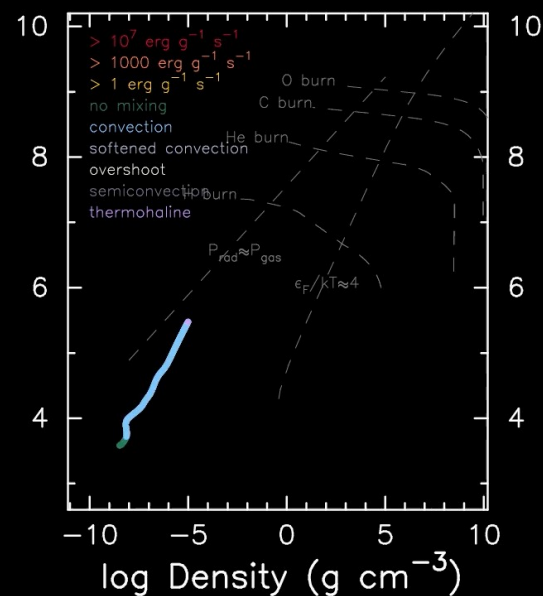
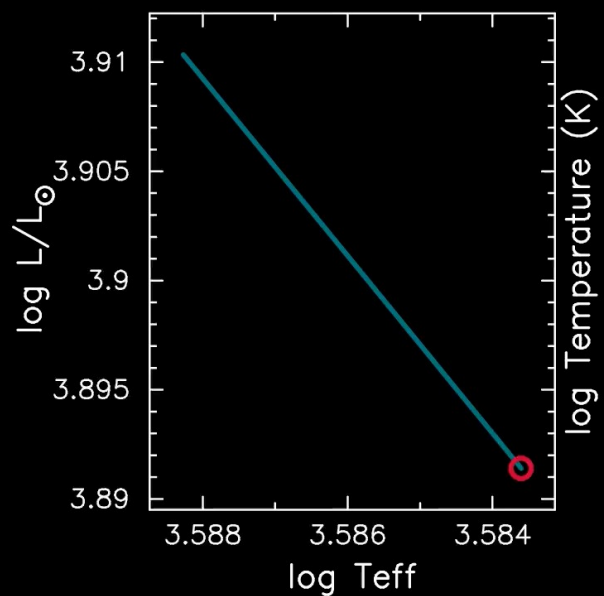
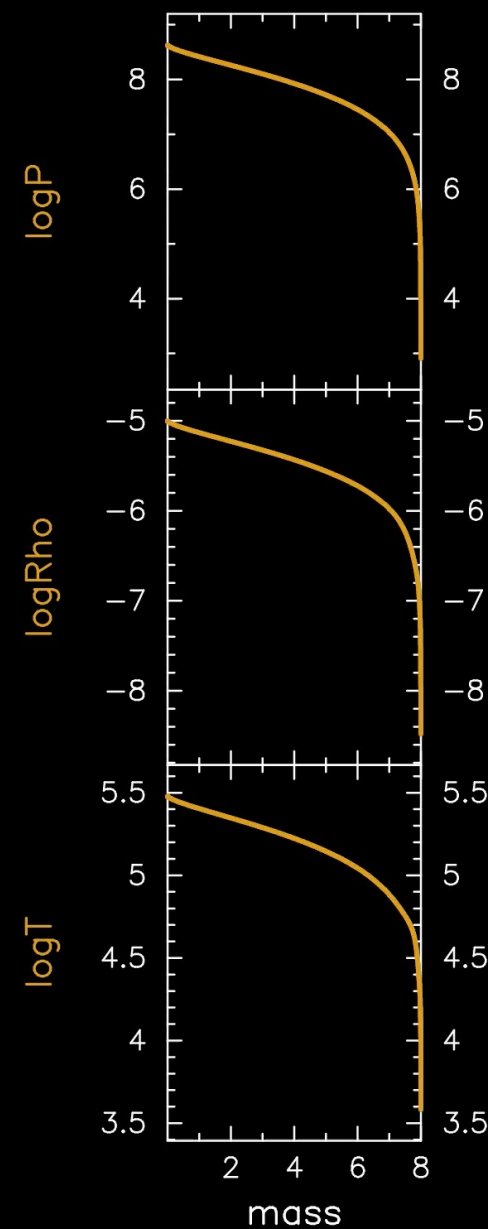
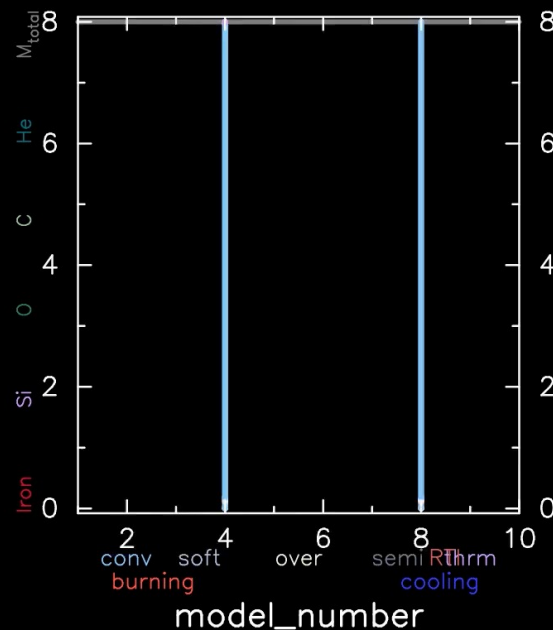
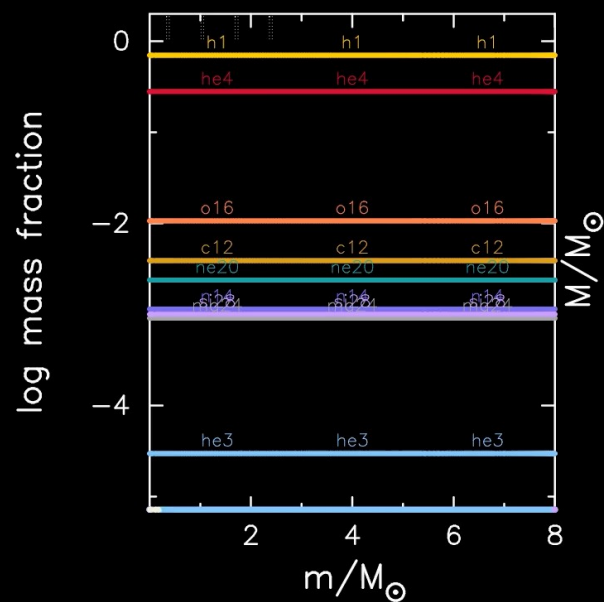
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age 2.595868e-4 yrs

model 10

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model 10

